17/02/24 (FN)

Hall Ticket Number:

MR-21

Code: B314CB6S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, Jan/Feb-2024

Computer Organization and Architecture

(Common to CSB, CSM & CSD)

Time: 3 Hours

Answer ALL Questions

Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define computer organization and computer architecture?	2	3	1	1
1. b)	What is one address, two address and three address instruction formats?	2	2	1	1
1. c)	What are the functions of control memory?	2	2	2	2
1. d)	What is Register Indirect Addressing mode? Give an example?	2	2	2	2
1. e)	Solve (1001101 - 10101001) using 2's complement.	2	3	3	2
1. f)	Add given signed decimal number +86 and -17. Perform (using 8 – bit representation) using 2's complement Addition.	2	3	3	2
1. g)	What is interrupt? Give the steps for handling interrupt?	2	2	4	1
1. h)	What is Auxiliary memory?	2	2	4	3
1. i)	Compare RISC and CISC?	2	2	5	2
1. j)	Explain the cache incoherence.	2	2	5	3

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the components of the Computer system.	5	2	1	1
2. b)	List and explain the steps involved in the execution of a complete instruction.	5	2	1	3
	OR				
2. c)	What is Micro operation? Briefly explain the arithmetic micro operations?	5	2	1	1
2. d)	With neat sketch explain the process of transferring the data between various registers of computer using bus structures.	5	2	1	3
3. a)	Explain the three basic types of data manipulation instructions.	5	2	2	1
3. b)	Mention the advantages and disadvantages of microprogrammed control hardwired control.	5	2	2	2
	OR				
3. c)	What do you mean by addressing mode? Explain the following addressing modes with examples. i) Index addressing mode ii) Relative addressing mode.	5	2	2	2
3. d)	Explain the block diagram of Microprogrammed control unit?	5	2	2	2
4. a)	Discuss about fixed point and floating-point representations.	5	2	3	2
4. b)	Perform and explain arithmetic addition, subtraction, and overflow detection using fixed point representation.	5	3	3	2
	OR				
4. c)	Perform floating point addition using the numbers $0.256X10^4$ and $0.4375X10^3$ use the floating point addition algorithm.	5	3	3	2
4. d)	Explain the multiplication of positive numbers using a booth multiplier with a neat sketch.	5	2	3	2

P.T.O.

5. a)	Explain the READ and WRITE operations in Associative Memory.	5	2	4	3
5. b)	Draw the block diagram of a DMA controller and explain its functioning?	5	2	4	3
	OR				
5. c)	Explain in detail parallel priority interrupt hardware?	5	2	4	1
5. d)	Explain direct mapping technique in cache memory?	5	2	4	2
6. a)	Explain the instruction pipeline processing in RISC architecture.	5	2	5	3
6. b)	Elucidate array processor in detail.	5	2	5	2
	OR				
6. c)	What are the various forms available for establishing an interconnection network in a multi-processor system?	5	2	5	1
6. d)	Does pipelining get affected by data dependencies among the instruction? Justify your answer with lucid examples.	5	2	5	3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome